Lab 5

Part 1

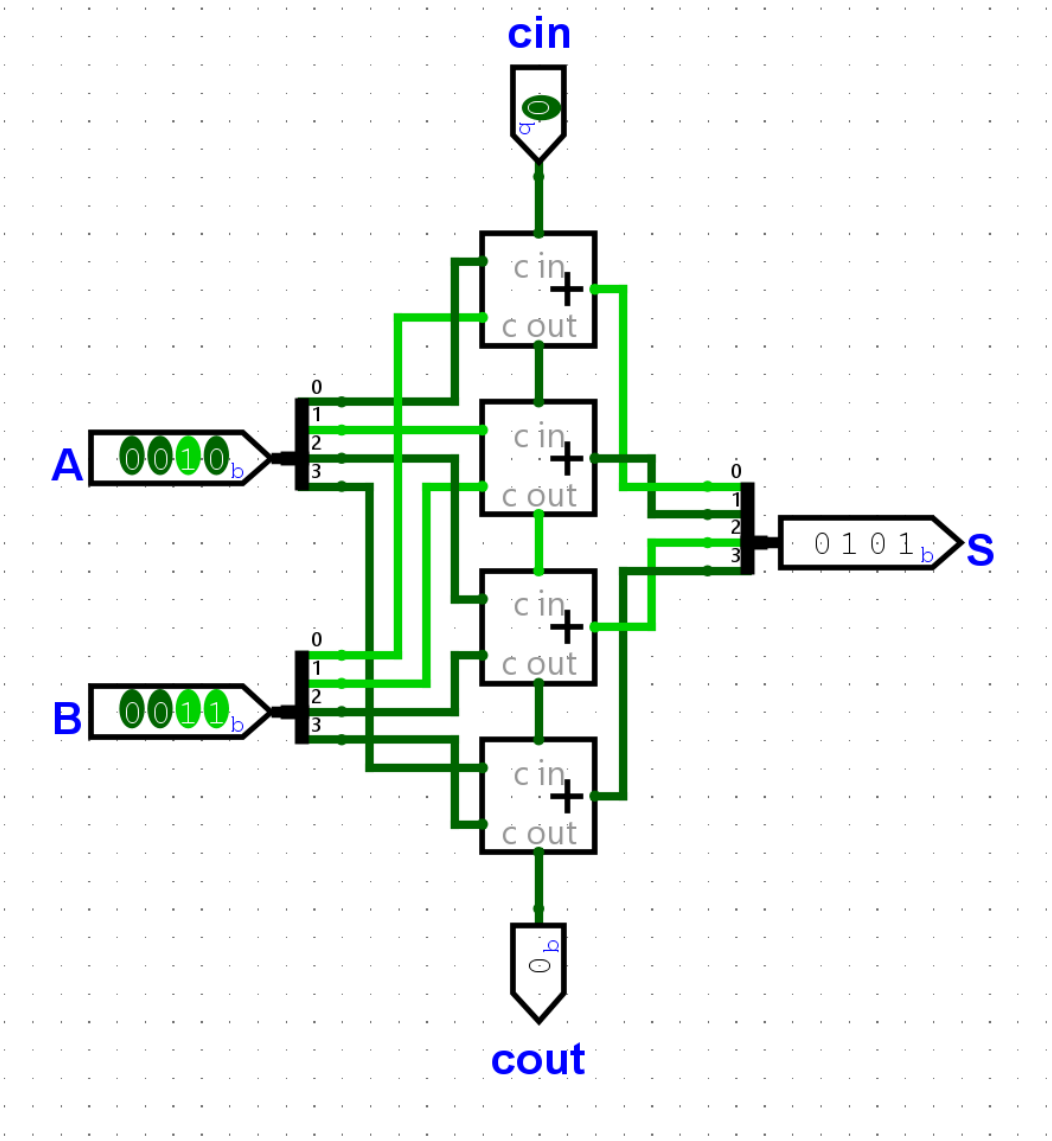


Figure 1 - 4 bit adder

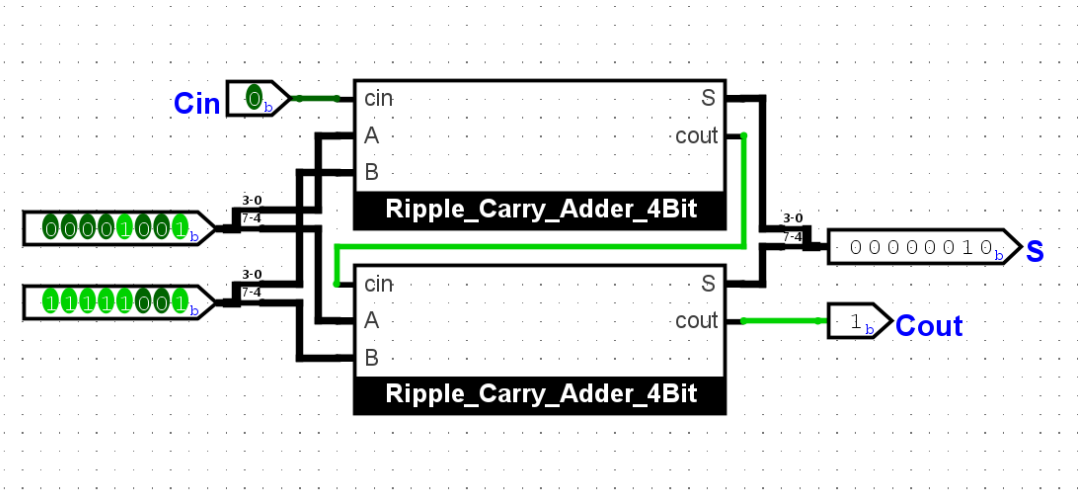


Figure 2 - Two 4bit adders to make 8bit adder

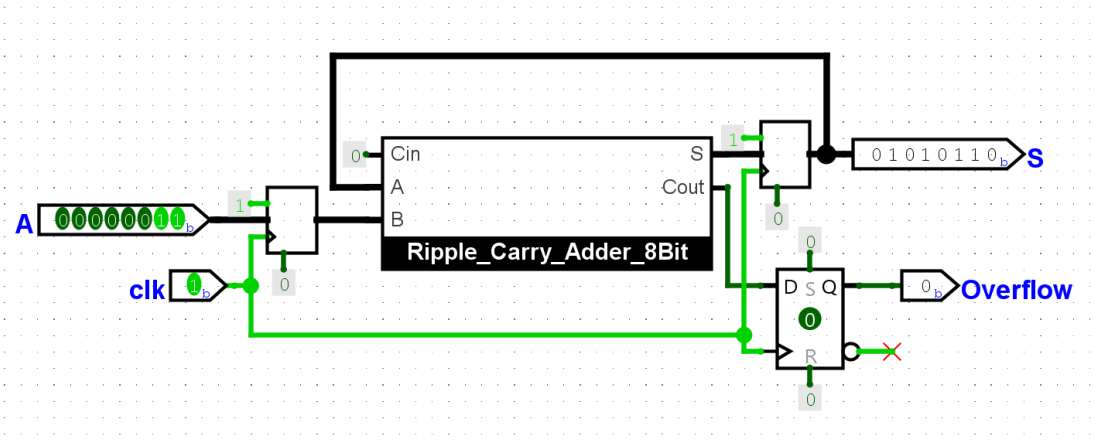


Figure 3 - The initial S state is undefined, which Logisim takes issue with since undefined can't be used for addition. I had to dig for a certain setting that changes initial states to 0s instead.

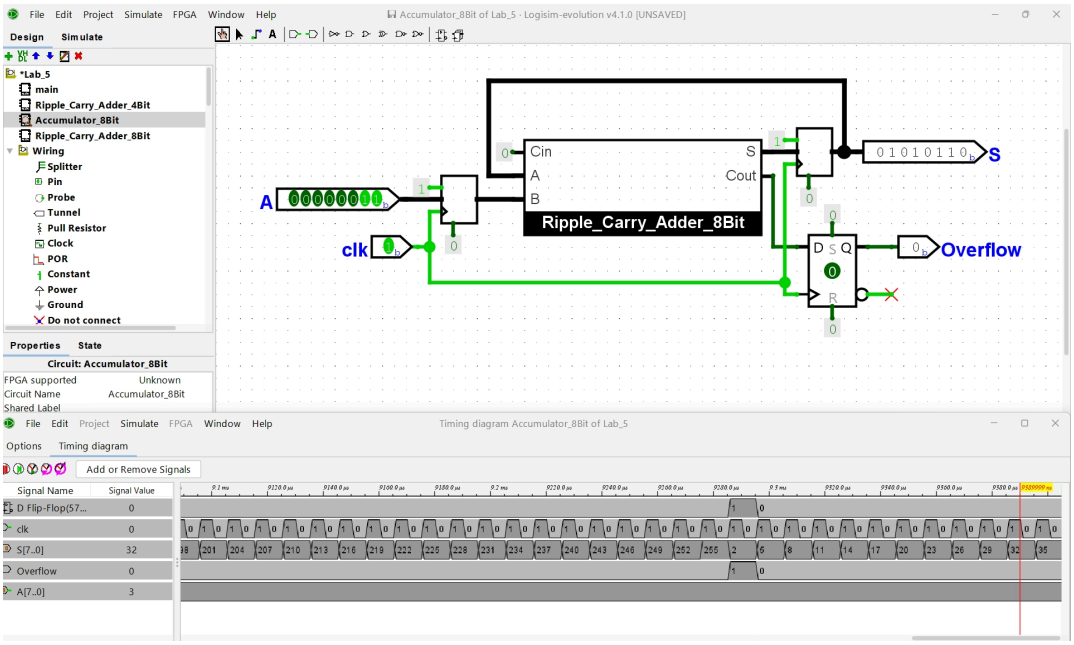


Figure 4 - Timing diagram of Accumulator

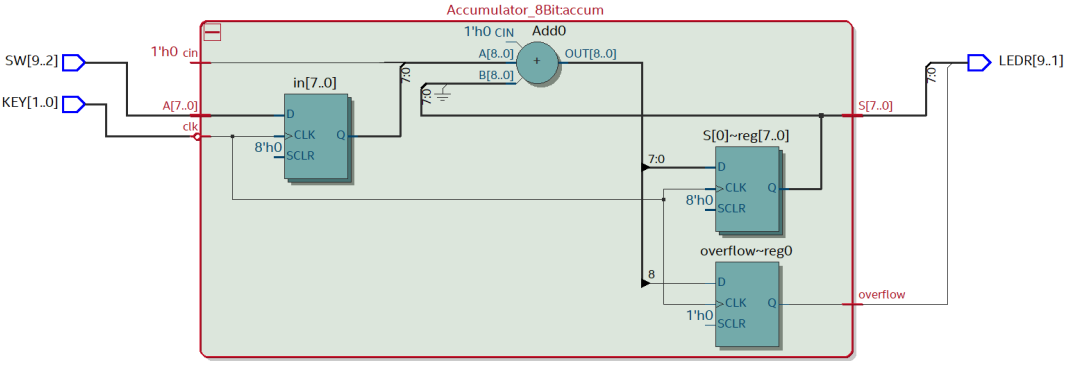


Figure 5 - Quartus RTL view of Accumulator

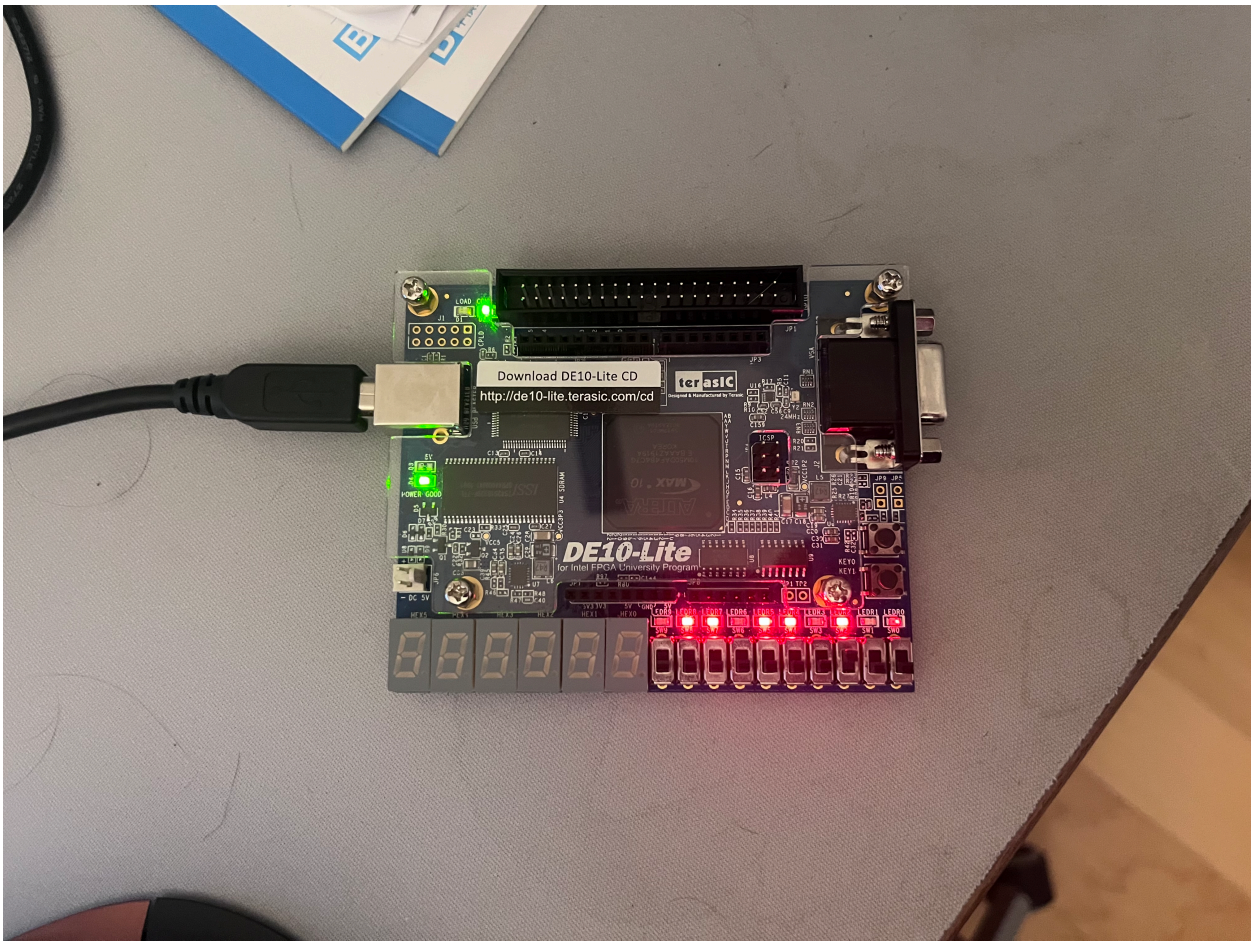


Figure 6 - Accumulator on FPGA

Part 2

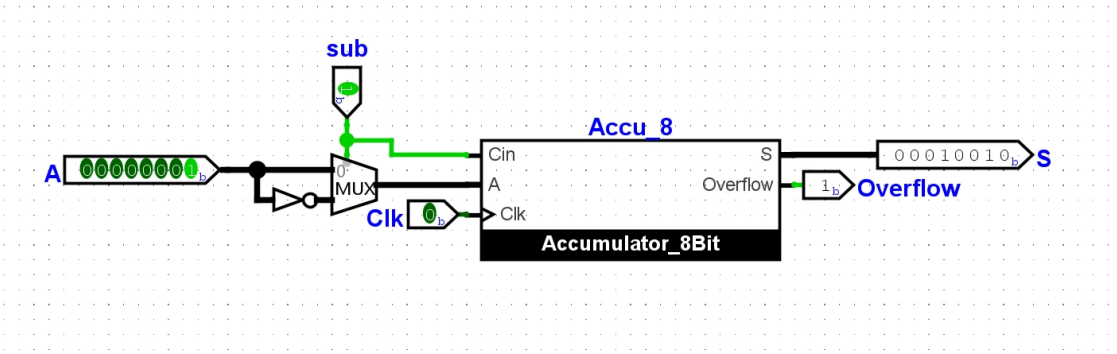


Figure 7 - Subtracting and adding accumulator with MUX

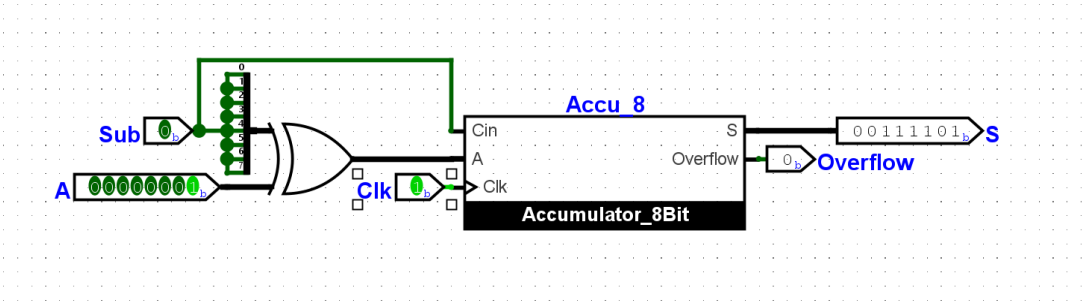


Figure 8 - Subtracting and adding accumulator with XOR

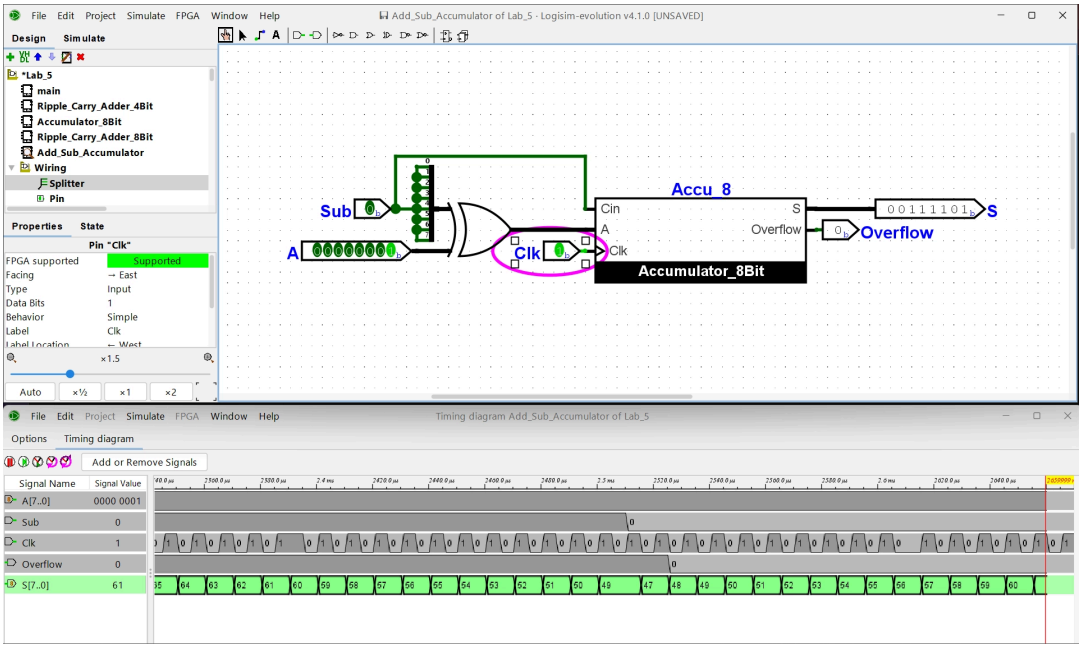


Figure 9 - When switching from subtraction to addition and vice versa, we skip a value. Not sure why yet.

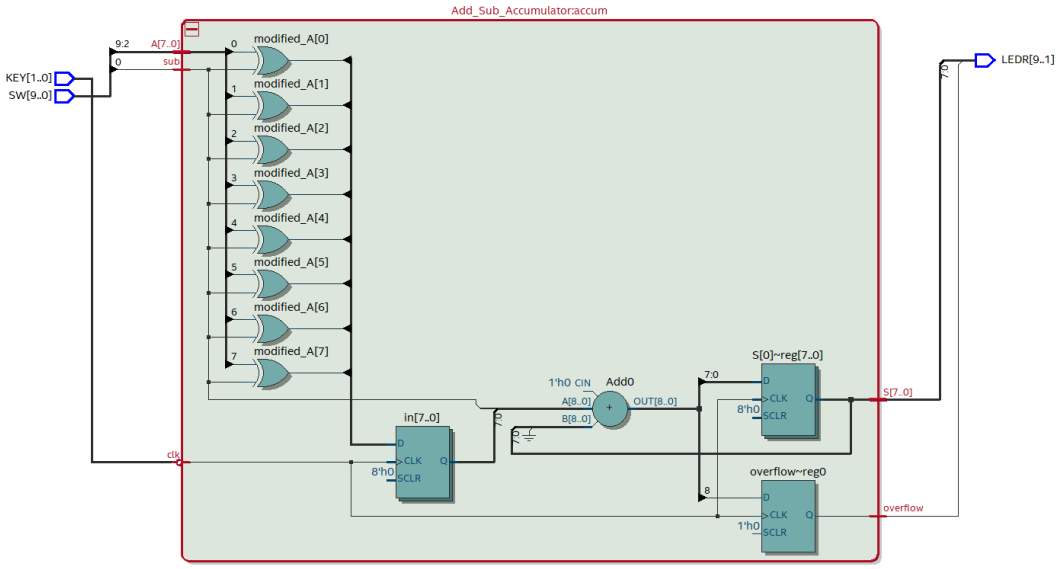


Figure 10 - RTL view showing XOR gates.

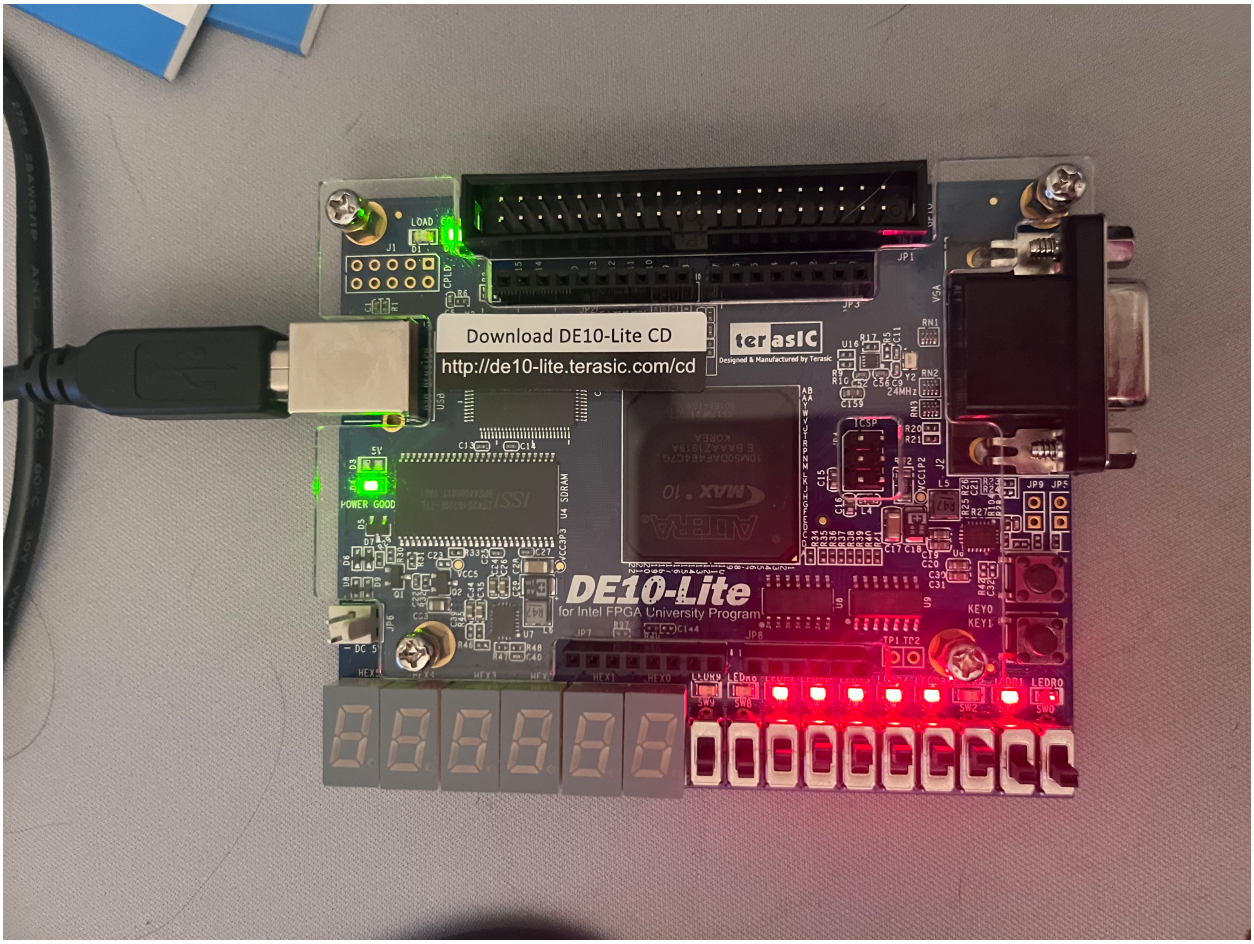


Figure 11 - AddSub Accumulator on FPGA

Part 3

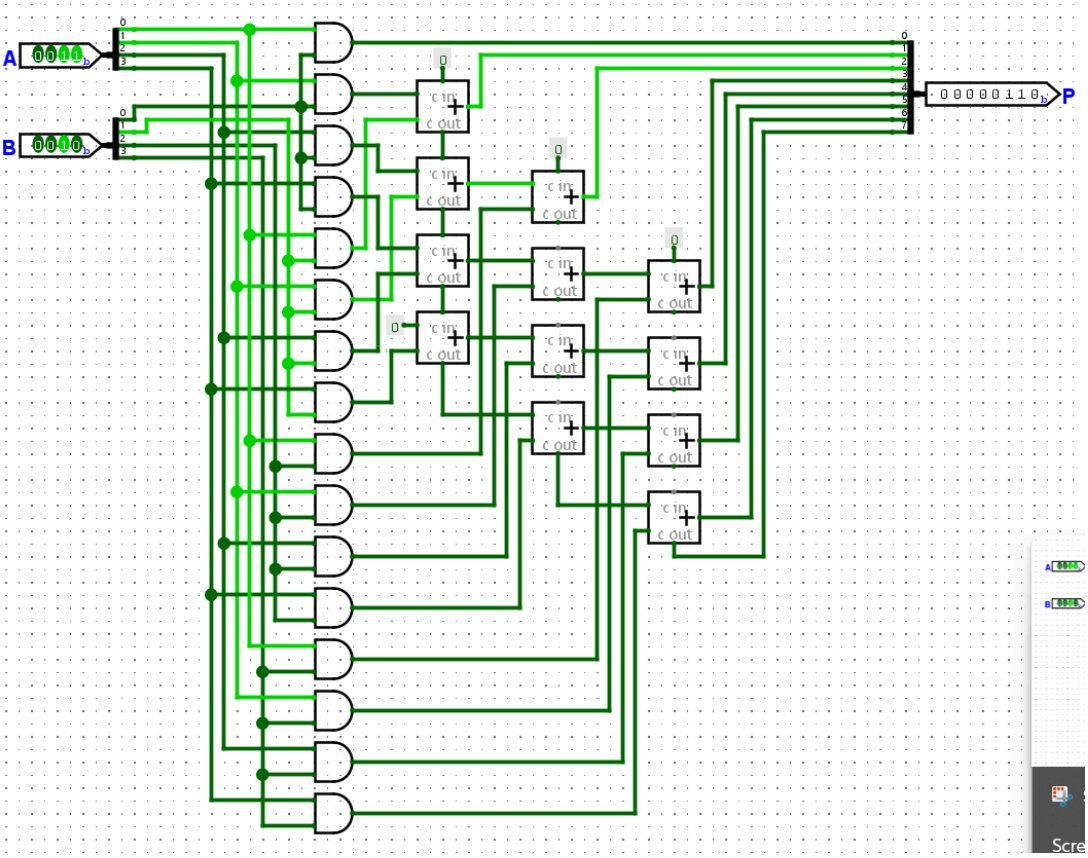


Figure 12 - 4bit multiplier circuit

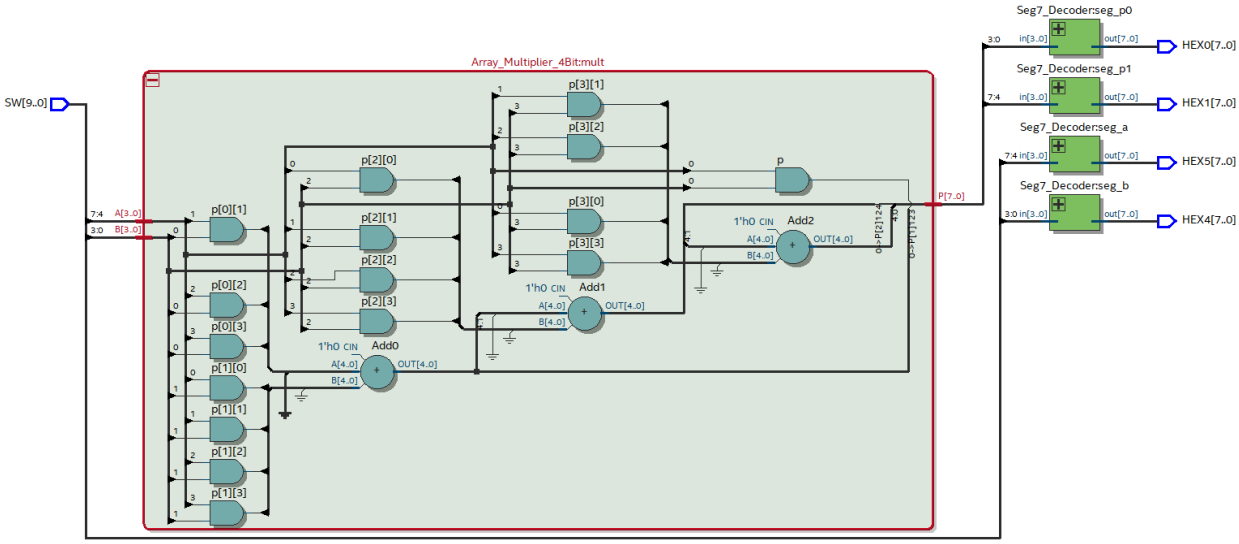


Figure 13 - 4bit multiplier circuit RTL view

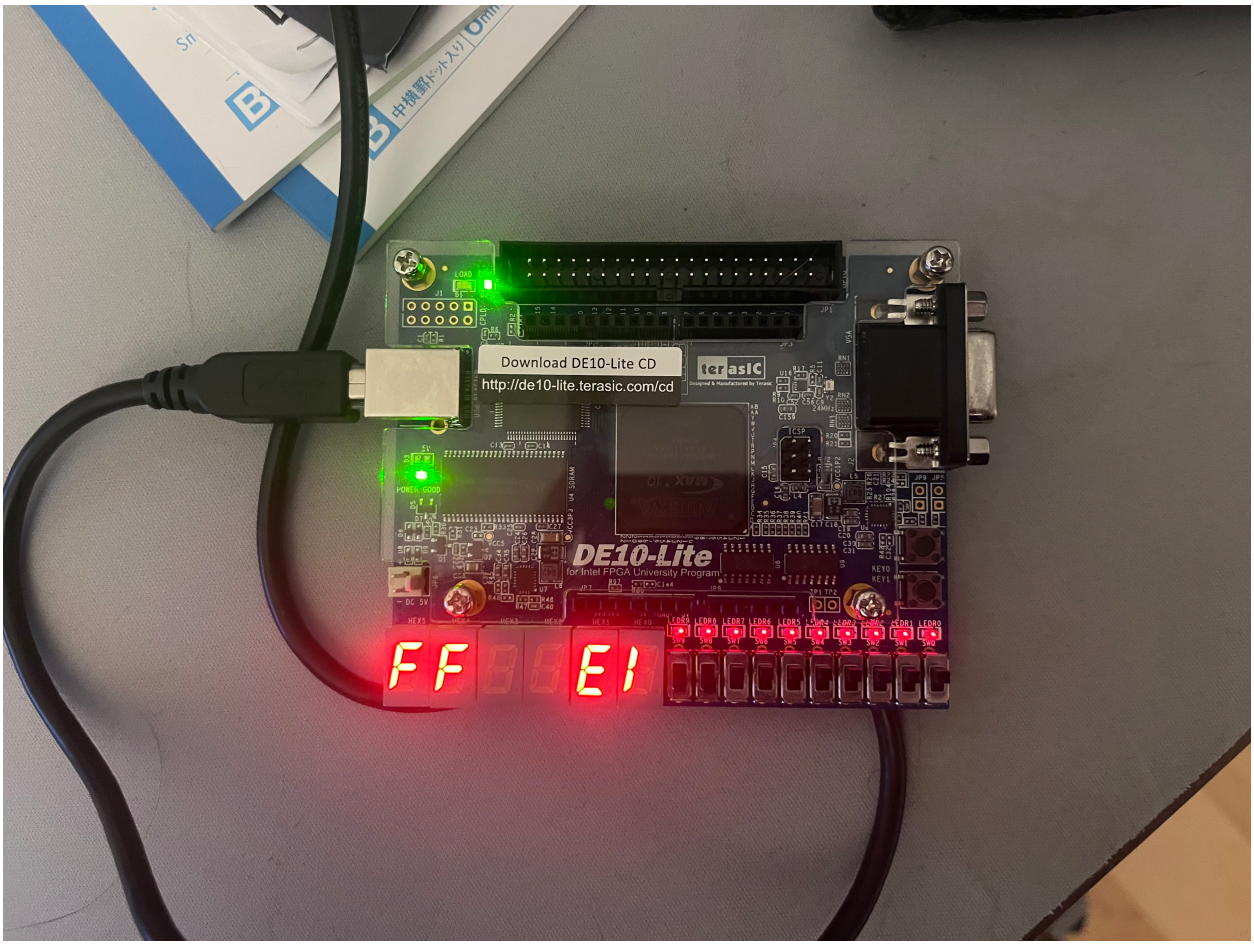


Figure 14 - 4bit Multiplier on FPGA with 2 single digit HEX inputs and 1 double digit HEX output.

Part 4

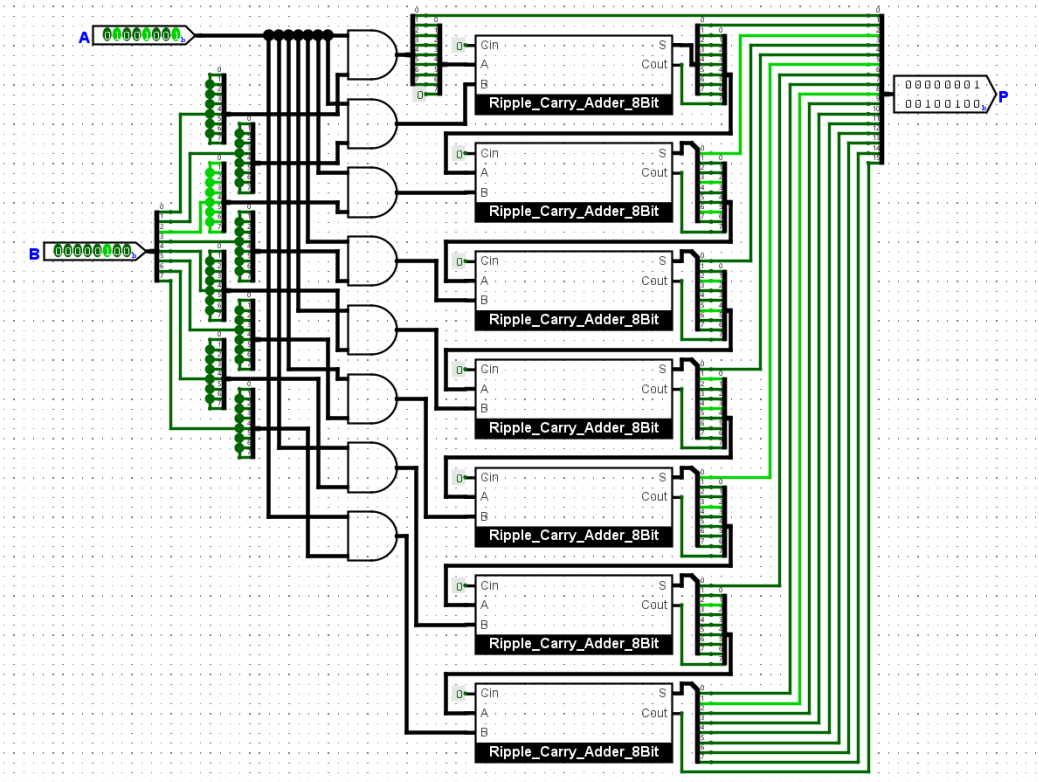


Figure 15 - Logisim 8bit multiplication circuit

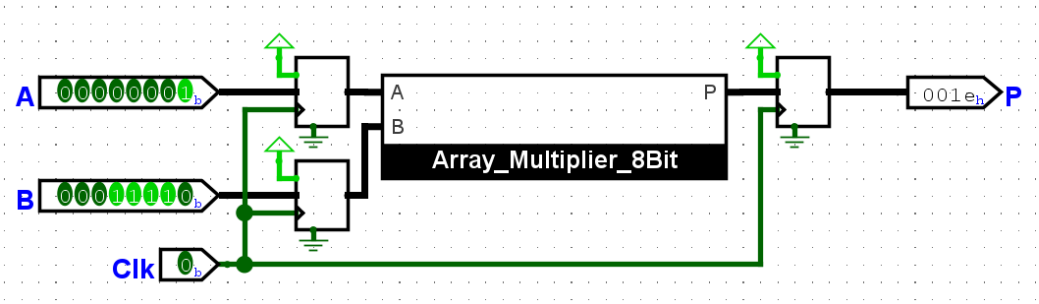


Figure 16 - Logisim Registered 8bit Multiplier

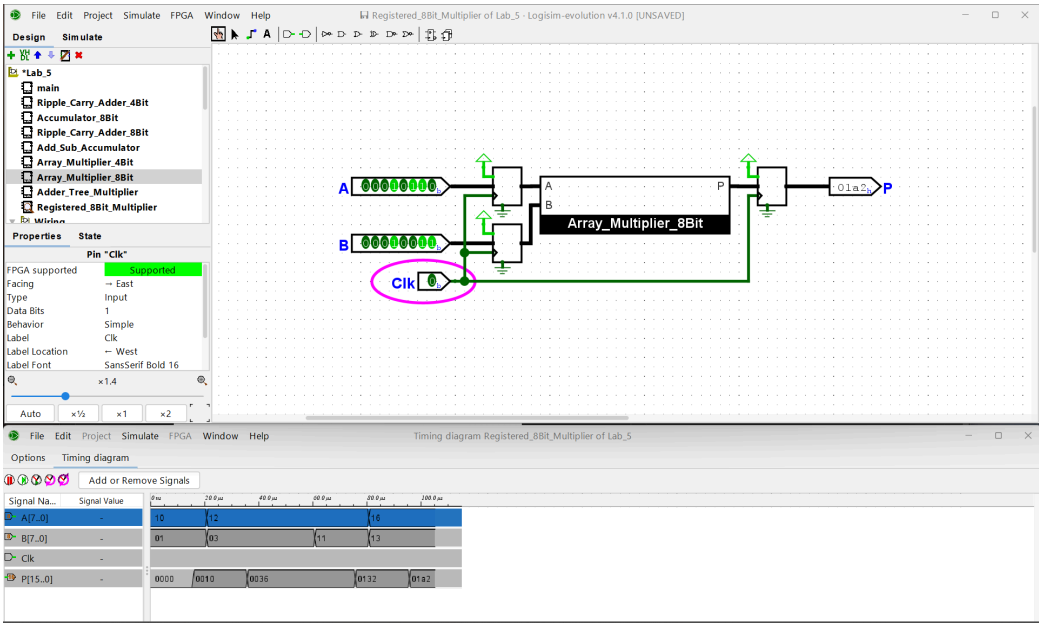


Figure 17 - Registered 8bit Multiplier Timing Diagram

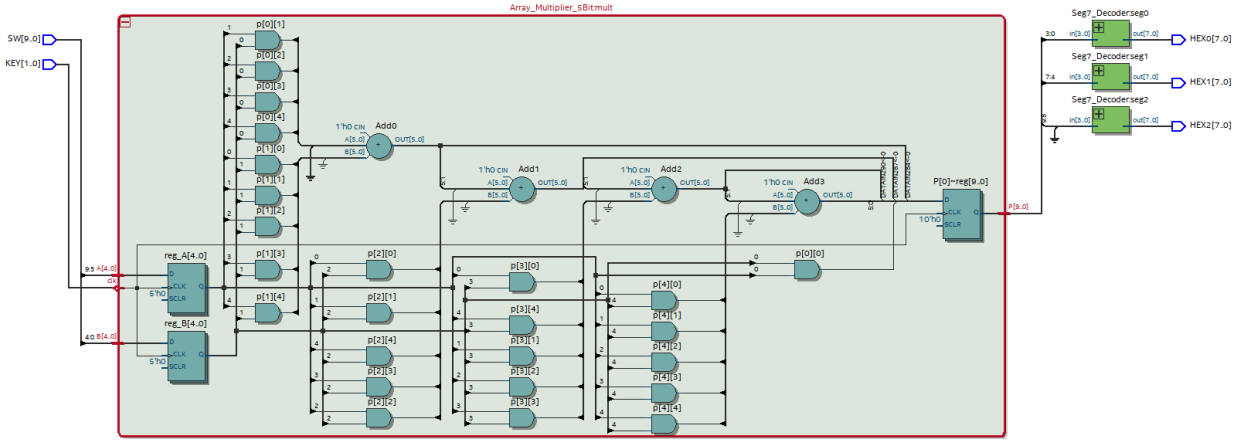


Figure 18 - Quartus RTL view 5bit multiplication circuit with registers.

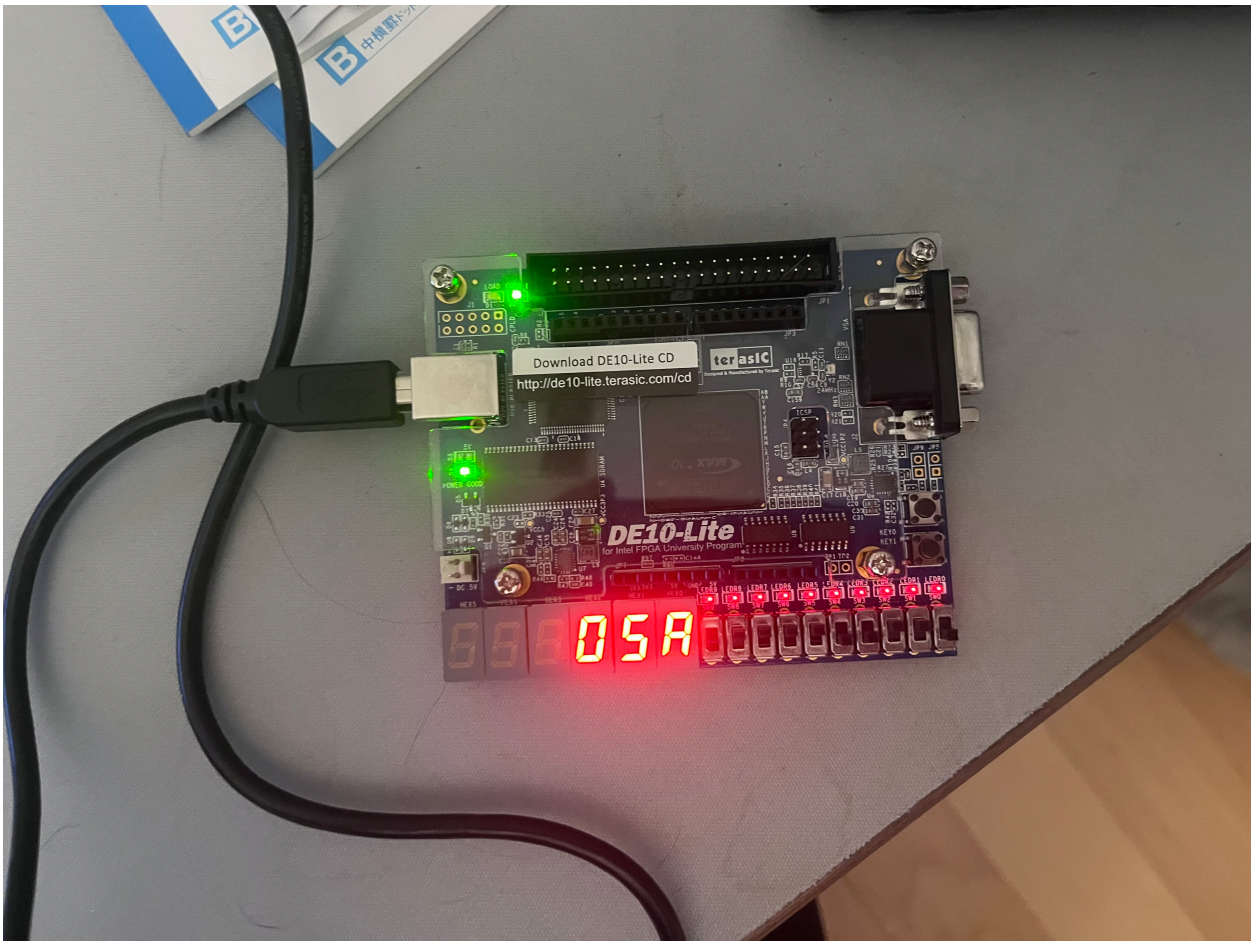


Figure 19 - 5bit Multiplication on FPGA

Part 5

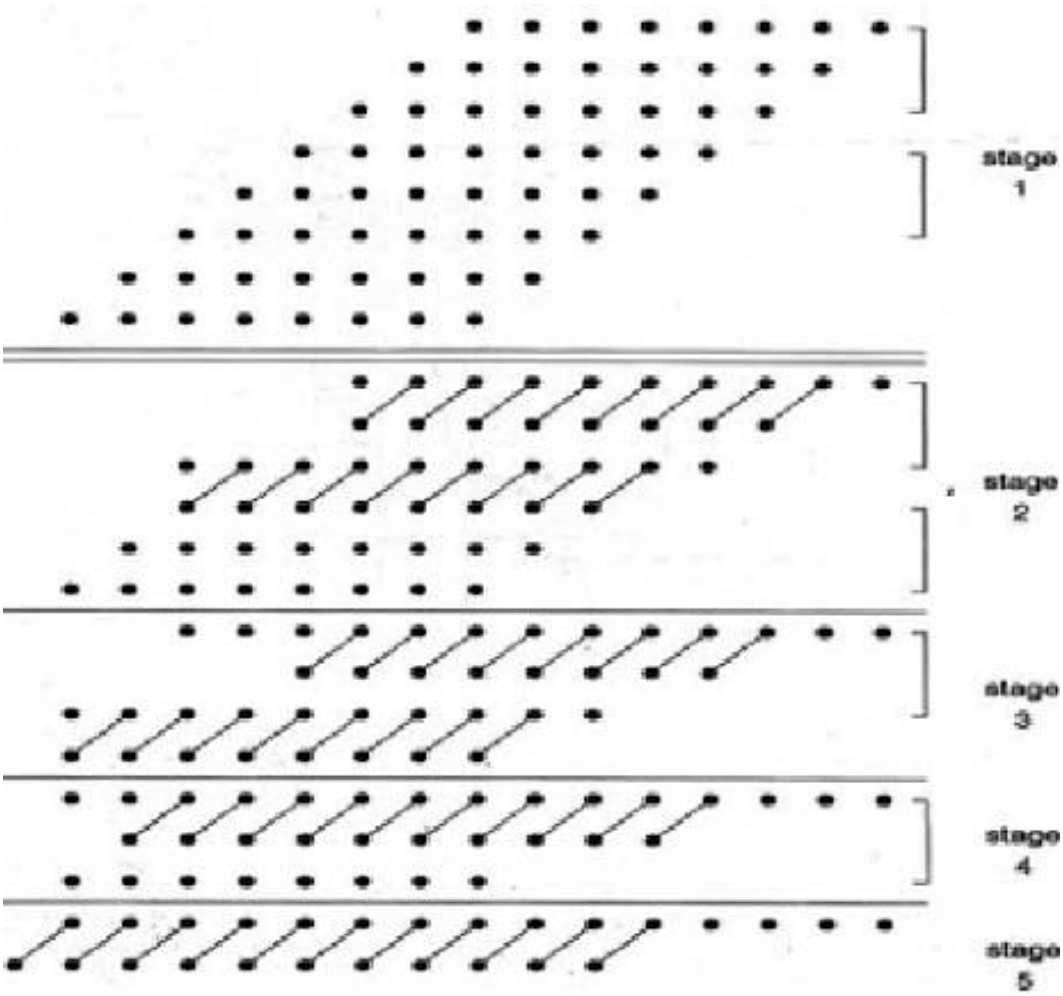


Figure 20 - Deva, Vasu & Hegadi, Rajendra. (2012). Design and Development of 8-Bits Fast Multiplier for Low Power Applications. International Journal of Engineering and Technology. 4. 774-780. 10.7763/IJET.2012.V4.482.

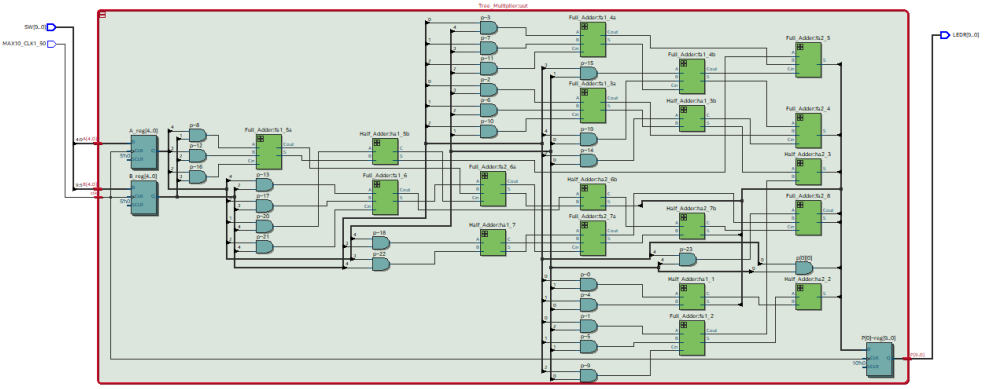


Figure 21 - RTL view of circuit. The previous figure is just an example of the method and it only uses half adders. This one uses full adders in addition.

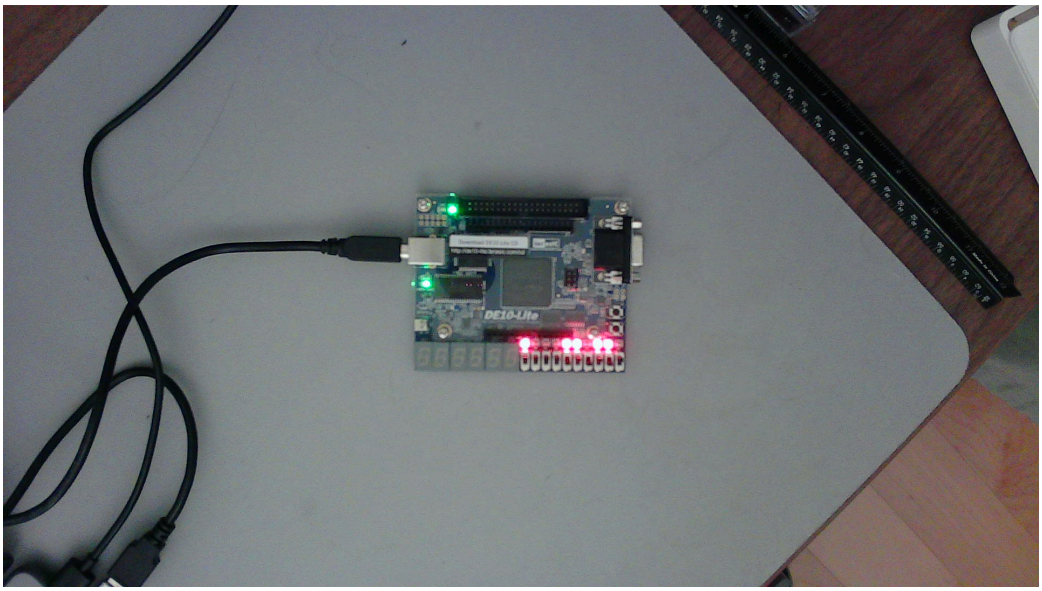


Figure 22 - Adder Tree Multiplier on FPGA